Name

Mathematics Extension 1 Year 12

Binomial Distributions

Book 1

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Syllabus Content

ME1-12-06 solves problems involving binomial distributions, sampling distribution of the mean and the central limit theorem

Bernoulli distributions

- Define a Bernoulli random variable as a model for two-outcome situations, referred to as success and failure
- Model a Bernoulli trial using a random variable, X, where the probability of success, $X = 1$, is p and the probability of failure, $X = 0$, is $q = 1 - p$
- Define a Bernoulli distribution as the discrete probability distribution of a Bernoulli random variable
- Establish and use the formulae $\mathrm{E}(X) = \mu = p$ to find the mean, and $\mathrm{Var}(X) = p(1 - p)$ to find the variance of the Bernoulli distribution with parameter p
- Identify contexts suitable to be modelled by Bernoulli random variables
- Solve practical problems involving Bernoulli random variables

Binomial distributions

- Define a binomial random variable, X, as the number of 'successes' in n independent and identically distributed Bernoulli trials, with the same probability of success p in each trial, where n is a fixed finite number
- Identify contexts suitable to be modelled using binomial distributions
- Define a binomial experiment as a fixed number of Bernoulli trials
- Use the notation $X \sim \mathrm{Bin}(n, p)$ to represent a binomial random variable, X, where n is the number of trials and probability of success is p
- Recognise the significance of ${}^{n}C_{r}$ as the number of ways in which an outcome with r successes can occur in n trials
- Use the formula $P(X = r) = {}^{n}C_{r}p^{r}(1 - p)^{n-r}$ to find the probability of r successes in a binomial experiment involving n trials
- Calculate the expected frequencies of the various possible outcomes from a binomial experiment, where the expected frequency of an outcome happening r times in n trials is $n \times P(X = r)$
- Use the formulae $\mathrm{E}(X) = \mu = np$ to find the mean, and $\mathrm{Var}(X) = np(1 - p)$ to find the variance when $X \sim \mathrm{Bin}(n, p)$
- Solve practical problems involving binomial distributions and binomial probabilities with and without online computational applications, excluding the normal approximation to the binomial distribution

Counting Arrangements of Two Elements

Review

☐ Recall permutations

a How many ways to arrange the letters:

TOAST

b How many ways to arrange the letters:

COOGEE

c There are 4 blue and 8 red cups on a shelf. How many ways are there to arrange them?

☐ Recall combinations

a How many ways are there to choose 4 people out of 10 to form a committee?

b Write the combinations formula:

$${}^{n}C_{r} = \binom{n}{r} =$$

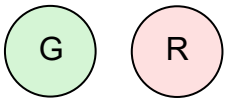
c There are 10 chairs in a row.
How many ways are there to choose 4 to paint red?

d Explain, using a combinatorial argument, why $\binom{12}{5} = \binom{12}{7}$

Investigation List arrangements of two elements

Imagine you have two colours of marbles.

Let the two colours be green (G) and red (R).



Let n represent the number of marbles. Let r be the number of green marbles (G).

List all possible marble arrangements for $n = 4, r = 2$ (there are 6)

GGRR GRGR ....

List all possible marble arrangements for $n = 5, r = 2$ (there are 10)

List all possible marble arrangements for $n = 5, r = 3$ (there are 10)

Now imagine you need to count the number of arrangements for $n = 10, r = 4$.

One possible arrangement is .....

To count all possible arrangements, we can use **permutations with identical elements**:

We can write this much more concisely using the **combinations formula**:

So, arranging 4 green and 6 red marbles in a row is mathematically equivalent to choosing 4 objects out of 10.

Imagine you have 10 numbered slots in a row. Focus entirely on placing the 4 green marbles.

To place them, you must choose **4 slots out of the available 10**.

Because the green marbles are identical, their order does not matter; swapping two green marbles changes nothing.

Once those 4 slots are chosen, the 6 identical red marbles are forced into the remaining 6 slots. There is only 1 way to do this.

Therefore, counting the arrangements is exactly equal to choosing the slots.

ⓘ Arrangements of Successes and Failures

Imagine you have n trials of an experiment.

The number of arrangements of r successes and $(n-r)$ failures is:

$$^nC_r = \binom{n}{r} = \frac{n!}{r!\,(n-r)!}$$

We can use this formula if we have **binary** (two) outcomes with no restrictions.

- Successes and failures
- Yes or no
- Green and red
- Heads and tails, etc.

ⓘ **Interpret** whether ^nC_r is appropriate

Identify whether you can use ^nC_r to count how many ways to:

a	arrange 5 green marbles and 7 red marbles in a line.	**b**	arrange 5 green marbles, 7 red marbles, and an orange marble in a line.
c	arrange the letters Y Y N N N	**d**	choose 3 students out of 9 for a committee
e	5 unpainted marbles in a line. How many ways to paint 3 red.	**f**	Number of strings of length 5 made from 1s and 0s containing exactly two 1s.
g	choose 3 students out of 9 for a committee, then assigning them roles of president, treasurer, secretary	**h**	how many ways there are to get 2 heads and 3 tails in five coin flips

1

a List every arrangement of 3 successes and 1 failure in 4 trials.

SSSF

SSFS

SFSS

FSSS

b How many are there?

4

c Is this the same as $\binom{4}{1}$?

Yes

2 Without calculating either, state which of ${}^{8}C_{2}$ and ${}^{8}C_{6}$ is larger. Justify your answer.

They are equal by the identity $\binom{n}{r} = \binom{n}{n-r}$

3 In how many orders can 2 successes and 4 failures occur in 6 trials?

15

4 In how many orders can 4 successes and 3 failures occur in 7 trials?

35

5 How many ways are there to get exactly 6 heads in 9 coin flips, with the rest being tails?

84

6 In how many orders can 3 successes and 5 failures occur in 8 trials, if the **first** trial is a failure?

$\binom{7}{3} = 35$

7 In how many orders can 3 successes and 5 failures occur in 8 trials, if **no two successes are adjacent**?

Write the 5 failures first: $_F_F_F_F_F_$.

This creates 6 gaps. Choosing 3 different gaps for the 3 successes guarantees no two are adjacent. ${}^{6}C_{3} = 20$.

8 A coin is tossed four times and the result is recorded. Janice wins if there are exactly two heads.

a List all the ways that this could occur, that is, in any order.

HHTT, HTHT, HTTH, THHT, THTH, TTHH.

b By counting, verify that there are exactly 6 cases where Janice wins and find the probability of this outcome.

There are 6 winning outcomes from $2^4 = 16$ equally likely outcomes.

$$P(\text{win}) = \frac{6}{16} = \frac{3}{8}$$

c Explain why this is the same as the number of ways of ordering the word HHTT, and use combinatorics to count how many such arrangements exist. Does your answer agree with part **a**?

Number of arrangements of HHTT:

$$4C2 = \frac{4!}{2!\,2!} = 6$$

This agrees with part a.

d Show that this is equivalent to choosing two of the coins and placing them heads up, while placing the other two coins tails up.

Choose the two toss positions occupied by heads:

$$\binom{4}{2} = 6$$

Bernoulli Random Variables

Review

☐ Recall complementary probabilities

a $P(A) = 0.4$
What is $P(\text{not A})$?

b A spinner lands on red with probability 0.6.
What is $P(\text{not red})$?

c A random variable only takes on values 0 and 1. $P(X = 1) = 0.2$. What is $P(X = 0)$?

d If $P(a) = 0.3$, what is $P(\bar{a})$?

☐ Complete probability tables and find $E(X)$, $E(X^2)$

a

x	0	1
$P(X = x)$		0.4

$E(X) = \Sigma xP(x) =$ .....

$E(X^2) = \Sigma x^2P(x) =$ .....

b

x	1	2
$P(X = x)$	0.6	

$E(X) = \Sigma xP(x) =$ .....

$E(X^2) = \Sigma x^2P(x) =$ .....

ⓘ Bernoulli Trial

A situation is a Bernoulli trial when there are **exactly 2** possible outcomes.

ⓘ **Identify** Bernoulli trials

✗	✓	✓	✗
A die is rolled. The number showing is recorded.	A die is rolled. It is recorded whether the number showing is 6.	A biased coin with P(head) = 0.99 is tossed. It is recorded whether it shows a head.	A biased coin is tossed 20 times. The number of heads is recorded.

Decide whether these are Bernoulli trials.

a	A patient is given a new drug. It is recorded whether their blood pressure falls.	**b**	One card is drawn from a deck. It is recorded whether the card is a heart.
c	A single traffic light is observed once. It is recorded whether the light is green.	**d**	A single traffic light is observed once. Its colour, red, yellow or green, is recorded.
e	One student is chosen at random from a school. Their year group is recorded.	**f**	One seed is planted. It is recorded whether the seed germinates.
g	A basketballer takes 10 free throws. The number scored is recorded.	**h**	A voter is asked whether they voted ‘yes’ in a referendum.

ⓘ Bernoulli Trial

A situation is a Bernoulli trial when there are **exactly 2** possible outcomes.

We model the two outcomes as either a 'success' or a 'failure' and give them values 0 and 1.

Let $X = \{0,1\}$ be a Bernoulli random variable.

$P(X = 0)$: probability of failure.

$P(X = 1)$: probability of success.

'Success' and 'failure' are defined by the question.

ⓘ **Interpret** Bernoulli trials

State in words what a success is and then write $P(X = 1)$.

X represents whether a light is green.

$X = 1$ if the light is green.

A light is green with probability 0.6.

Success: the light is green.

$P(X = 1) = 0.6$

X represents whether a light is green.

$X = 1$ if the light is not green.

A light is green with probability 0.6.

Success: the light is not green.

$P(X = 1) = 0.4$

a A component is faulty in 15% of cases. Let $X = 1$ if the component is faulty

Success: .....

$P(X = 1) =$ .....

b A seed germinates with probability 0.85. $X = 1$ if the seed germinates.

c A seed germinates with probability 0.85. $X = 1$ if the seed fails to germinate.

d A machine part passes inspection 92% of the time. $X = 1$ if the part fails inspection.

e A biased die shows a six with probability $\frac{1}{5}$. $X = 1$ if the die shows a six.

f A biased die shows a six with probability $\frac{1}{5}$. $X = 1$ if the die does not show a six.

ⓘ Bernoulli Random Variables

A situation is a Bernoulli trial when there are **exactly 2** possible outcomes.

We model the two outcomes as either a 'success' or a 'failure' and give them values 0 and 1.

Let the probability of success be p, the probability of failure $1-p$.

$$P(X=x)=\begin{cases}p & x=1\\ 1-p & x=0\end{cases}$$

or using a table:

x	0	1
$P(x)$	$1-p$	p

A Bernoulli random variable is defined only by probability of success p.

We can write

$$X\sim\text{Bernoulli}(p)$$

to say 'X is a Bernoulli random variable with probability of success p.'

ⓘ **Interpret** Bernoulli random variables

Let X be whether you roll a 6 on a die.

$$P(X=x)=\begin{cases}\frac{1}{6} & x=1\\ \frac{5}{6} & x=0\end{cases}$$

x	0	1
$P(x)$	$\frac{5}{6}$	$\frac{1}{6}$

$X\sim\text{Bernoulli}\left(\frac{1}{6}\right)$

Success $X=1$ is: .....

Failure $X=0$ is: .....

The probability of success $P(X=1)$ is .....

The probability of failure $P(X=0)$ is .....

$X\sim\text{Bernoulli}\left(\frac{1}{6}\right)$ means 'X is a random variable with only ..... outcomes, s..... or f..... The probability of success is .....'

Write the piecewise function, table of values, and $X\sim$ notation for these Bernoulli scenarios

a Let X be whether you flip a heads on a fair coin.

b Let X be whether you choose a vowel (A, E, I, O, U) from letters of the alphabet.

Expected Value and Variance of Bernoulli Distributions

A Bernoulli distribution shows the probability of 'success' and 'failure.'

Calculate expected value using the same formulas as discrete random variables.

Expected Value	Variance & Standard Deviation
$E(X) = \Sigma x P(x)$	$\text{Var}(X) = [\Sigma x^2 P(x)] - E(X)^2$ $\sigma(X) = \sqrt{\text{Var}(X)}$

Example Calculate expected value and variance for a Bernoulli distribution

A biased coin shows heads with probability 0.3.

Let $X = 1$ if the coin shows heads.

Construct the Bernoulli distribution table and find $E(X)$, $\text{Var}(X)$, and $\sigma(X)$.

x	0	1
$P(x)$	0.7	0.3

$$E(X) = \Sigma x P(x) = 0(0.7) + 1(0.3)$$
$$= 0.3$$

$$\text{Var}(X) = \Sigma x^2 P(x) - E(X)^2$$
$$= 0^2(0.7) + 1^2(0.3) - (0.3)^2$$
$$= 0.21$$

$$\sigma(X) = \sqrt{0.21} \approx 0.458$$

Guided Practice

a A biased coin shows heads with probability 0.8.
Let $X = 1$ if the coin shows heads. Find $E(X)$, $\text{Var}(X)$ and $\sigma(X)$.

$E(X) = 0.8$
$\text{Var}(X) = 0.16$
$\sigma(X) = 0.4$

b Consider the general case for a Bernoulli distribution where $P(X = 1) = p$.
Find $E(X)$, $\text{Var}(X)$, and $\sigma(X)$.

$E(X) = p$
$\text{Var}(X) = p(1-p)$
$\sigma(X) = \sqrt{p(1-p)}$

Expected Value and Variance of Bernoulli Distributions

Expected Value	Variance & Standard Deviation
$E(X) = \Sigma x P(x) = p$	$\text{Var}(X) = p(1-p)$ $\sigma(X) = \sqrt{p(1-p)}$

1 State whether each statement describes a Bernoulli random variable.

a Selecting at random an odd-numbered card from a set of numbered cards

Yes: odd or not odd.

b Choosing at random a red playing card from a deck of cards

Yes: red or not red.

c The number of tails when tossing 2 coins

No: the values are $0,1,2$.

d The colour of a traffic light

No: there are more than 2 possible colours.

e A missile hitting a target

Yes: hit or miss.

f Randomly selecting a boy from a school class

Yes: boy or not boy.

g A film reviewer's star rating of a film

No: there are more than 2 possible ratings.

h Selecting a faulty phone in a random quality-control check

Yes: faulty or not faulty.

i The age of a person selected at random in a shopping centre

No: age has more than 2 possible values.

j Guessing the correct answer to a true/false question

Yes: correct or incorrect.

2 Write each Bernoulli distribution as a piecewise function. Define the success condition.

a Passing a test if the probability of a pass is $\frac{7}{9}$

Let $X = 1$ for a pass.

$$P(X = x) = \begin{cases} \frac{7}{9} & \text{for } x = 1 \\ \frac{2}{9} & \text{for } x = 0 \end{cases}$$

b An arrow hitting a target with probability 84%

Let $X = 1$ for a hit.

$$P(X = x) = \begin{cases} 0.84 & \text{for } x = 1 \\ 0.16 & \text{for } x = 0 \end{cases}$$

c Getting a head when tossing a coin

Let $X = 1$ for a head.

$$P(X = x) = \begin{cases} \frac{1}{2} & \text{for } x = 1 \\ \frac{1}{2} & \text{for } x = 0 \end{cases}$$

d Winning a raffle in which there are 150 tickets sold

Let $X = 1$ for a win.

$$P(X = x) = \begin{cases} \frac{1}{150} & \text{for } x = 1 \\ \frac{149}{150} & \text{for } x = 0 \end{cases}$$

3 Draw a Bernoulli distribution table for each event.

a Selecting a girl at random from a group of 12 boys and 14 girls

$P(X=1)=\frac{14}{26}=\frac{7}{13}$.

X	0	1
$P(X=x)$	$\frac{6}{13}$	$\frac{7}{13}$

b Selecting an ace at random from a deck of cards

$P(X=1)=\frac{4}{52}=\frac{1}{13}$.

X	0	1
$P(X=x)$	$\frac{12}{13}$	$\frac{1}{13}$

c Getting a total of 5 when rolling 2 dice

favourable outcomes: $(1,4),(2,3),(3,2),(4,1)$.

$P(X=1)=\frac{4}{36}=\frac{1}{9}$.

X	0	1
$P(X=x)$	$\frac{8}{9}$	$\frac{1}{9}$

d Selecting a white card at random from a set of 10 white and 15 red cards

$P(X=1)=\frac{10}{25}=\frac{2}{5}$.

X	0	1
$P(X=x)$	$\frac{3}{5}$	$\frac{2}{5}$

e Selecting a number divisible by 3 when randomly selecting a card from 30 cards, each numbered with a different number from 1 to 30

There are 10 multiples of 3 from 1 to 30.

$P(X=1)=\frac{10}{30}=\frac{1}{3}$.

X	0	1
$P(X=x)$	$\frac{2}{3}$	$\frac{1}{3}$

4 On average, 8% of mail is junk mail. Find the mean and variance of a Bernoulli distribution for receiving junk mail.

$p=0.08$.

$E(X)=0.08$

$\text{Var}(X)=0.08(0.92)=0.0736$

5 A biased coin has probability $\frac{3}{4}$ of coming up heads.

a Write a Bernoulli distribution for tossing a head.

Let $X = 1$ for a head.

$$P(X = x) = \begin{cases} \frac{3}{4} & \text{for } x = 1 \\ \frac{1}{4} & \text{for } x = 0 \end{cases}$$

b For the distribution, find:

i the mean

$$E(X) = \frac{3}{4}$$

ii the variance

$$\text{Var}(X) = \frac{3}{4}\left(\frac{1}{4}\right) = \frac{3}{16}$$

6 For each Bernoulli distribution, find the mean and variance.

a $P(X = x) = \begin{cases} 0.7 & \text{for } x = 1 \\ 0.3 & \text{for } x = 0 \end{cases}$

$$E(X) = 0.7$$

$$\text{Var}(X) = 0.7(0.3) = 0.21$$

b $P(X = x) = \begin{cases} \frac{14}{15} & \text{for } x = 1 \\ \frac{1}{15} & \text{for } x = 0 \end{cases}$

$$E(X) = \frac{14}{15}$$

$$\text{Var}(X) = \frac{14}{15}\left(\frac{1}{15}\right) = \frac{14}{225}$$

c $p(x) = \frac{x+1}{3}$ for $x = 0,1$

$$E(X) = \frac{2}{3}$$

$$\text{Var}(X) = \frac{2}{3}\left(\frac{1}{3}\right) = \frac{2}{9}$$

d

X	0	1
$P(X = x)$	48%	52%

$$E(X) = 0.52$$

$$\text{Var}(X) = 0.52(0.48) = 0.2496$$

e

X	0	1
$P(X = x)$	$\frac{99}{100}$	$\frac{1}{100}$

$$E(X) = \frac{1}{100}$$

$$\text{Var}(X) = \frac{1}{100}\left(\frac{99}{100}\right) = \frac{99}{10000}$$

7 The mean of a Bernoulli distribution is $\frac{9}{10}$.

a Find the variance.

b Write the Bernoulli distribution as a table.

$p = \frac{9}{10}$.

$\text{Var}(X) = \frac{9}{10}\left(\frac{1}{10}\right) = \frac{9}{100}$

X	0	1
$P(X = x)$	$\frac{1}{10}$	$\frac{9}{10}$

8 The probability that a basketball player will shoot a basket is $\frac{9}{13}$. For the Bernoulli distribution for shooting a basket, find:

a the mean

b the variance

c the standard deviation

$E(X) = \frac{9}{13}$

$\text{Var}(X) = \frac{9}{13}\left(\frac{4}{13}\right) = \frac{36}{169}$

$\sigma = \sqrt{\frac{36}{169}} = \frac{6}{13}$

9 Find the Bernoulli distribution for getting a double when rolling 2 dice.
For this distribution, find:

a the mean

b the variance

c the exact standard deviation

There are 6 doubles among 36 outcomes.

$$P(X = x) = \begin{cases} \frac{1}{6} & \text{for } x = 1 \\ \frac{5}{6} & \text{for } x = 0 \end{cases}$$

$E(X) = \frac{1}{6}$

$\text{Var}(X) = \frac{1}{6}\left(\frac{5}{6}\right) = \frac{5}{36}$

$\sigma = \sqrt{\frac{5}{36}} = \frac{\sqrt{5}}{6}$

Binomial Random Variables

ⓘ Binomial Random Variables

A binomial random variable counts the number of 'successes' in repeated Bernoulli trials.

A binomial random variable requires these conditions:

1. The number of trials n is finite and fixed.
2. Each trial has exactly two outcomes: success or failure.
3. The trials are independent.
4. Probability p is the same in every trial.

ⓘ **Identify** whether number of trials is fixed

✗	A student guesses answers to a multiple-choice quiz **until she gets one correct**, and records how many questions she attempted.
✓	A student guesses answers to **all 20** questions, and records how many she gets correct.
✓	A die is rolled **60 times**; the number of sixes is recorded.
✓	**Eight** patients are given a new drug; the number who recover is recorded.
✗	A die is rolled **until the first six** appears; the number of rolls is recorded.

Determine whether the number of trials is fixed.

a	A coin is tossed 30 times.	**b**	A coin is tossed until a head appears.
c	A coin is tossed until a head appears, or 10 times, whichever comes first.	**d**	Twelve free throws are taken.
e	Free throws are taken until three are scored.	**f**	Every one of the 25 globes in a box is tested.

Identify whether each trial only has two outcomes

✗	A die is rolled 20 times and **the number shown** on each roll is recorded.
✓	A die is rolled 20 times and **whether each roll shows a six** is recorded.
✓	A 20-question quiz has four options per question. For each question we record **whether the answer is correct**.
✓	Fifteen voters each name one of six parties. For each voter we record **whether they name Party A**.
✗	Ten chess games are played. For each game we record **win, draw or loss**.

Determine whether each trial has two outcomes or not.

a	Record whether each of 20 globes is faulty.	**b**	Record the lifetime in hours of each of 20 globes.
c	Record whether each of 20 globes lasts more than 1000 hours.	**d**	Record the suit of each of 10 cards drawn.
e	Record whether each of 10 cards drawn is a heart.	**f**	Record whether each of 10 patients recovers fully, partially, or not at all.

Identify whether each trial is independent

✗	Six people **share a house**. We record whether each of the six catches the flu this winter.
✓	Six people **living in six different suburbs** are chosen. We record whether each catches the flu this winter.
✓	A card is drawn from a shuffled deck, its suit noted, and it is **replaced**. This is repeated 10 times. The number of hearts is recorded.
✓	Twenty adults are chosen at random from a population of five million and asked whether they support a proposal. ⚠ We treat this as independent as removing a few people from 5 million barely changes the probability.
✗	Twenty adults are chosen at random, but they are surveyed **in pairs, and each pair discusses the question before answering**.

Determine whether each trial is independent (or approximately independent).

a	Ten rolls of a die.	**b**	Ten cards drawn with replacement.
c	Ten cards drawn without replacement.	**d**	Whether each of 30 randomly chosen Australians owns a car.
e	Whether each of 30 students in one class passes a test they sat while allowed to talk to each other.	**f**	Whether each of 5 children in one family has blue eyes.

Identify whether p is the same in every trial

✗	An athlete takes 30 shots at goal. She **tires as she goes**, so each shot is a little less likely to score than the one before.
✓	An athlete takes 30 shots at goal, **resting fully between shots**. Each shot has probability 0.4 of scoring.
✓	A die is rolled 20 times; the number of sixes is recorded.
✓	A 20-question quiz has four options per question and is guessed throughout.
✗	A 20-question quiz is guessed throughout. **The first 10 questions have four options; the last 10 have five options.**

Determine whether each trial has the same probability

a	Twenty tosses of the same fair coin.	**b**	Twenty tosses, ten with a fair coin and ten with a biased coin.
c	Twenty cards drawn with replacement; record whether each is a heart.	**d**	Twenty cards drawn without replacement; record whether each is a heart.
e	A surgeon performs 40 operations over her career; she improves with experience.	**f**	A machine stamps 500 parts; it is serviced hourly and the fault rate stays at 2%.

ⓘ **Identify** whether a scenario can be modelled as a binomial random variable

A fair coin is tossed 40 times. The number of heads is recorded. Yes.	A die is rolled until the first six appears. The number of rolls is recorded. No. 'until the first six' means n not fixed.

Identify whether the context can be modelled as a binomial random variable.

a	3% of globes made by a machine are faulty. Twenty-five globes are chosen at random from 40 000. The number of faulty globes is recorded.	**b**	A die is rolled 20 times and the number shown on each roll is recorded.
c	A card is drawn from a shuffled deck, its suit noted, and it is replaced. This is repeated 12 times. The number of hearts is recorded.	**d**	An athlete takes 30 shots at goal. She tires as the session goes on, so each shot is less likely to score than the last. The number of goals is recorded.
e	Twenty people are chosen at random from a city of two million and asked whether they cycle to work. The number who say yes is recorded.	**f**	A student guesses every answer on a 15-question quiz. Each question has five options, one correct. The number of correct answers is recorded.
g	Free throws are taken until three have been scored. The number of throws is recorded.	**h**	A student guesses every answer on a 20-question quiz. The first 10 questions have four options; the last 10 have five. The number of correct answers is recorded.
i	Six housemates are recorded as catching influenza or not this winter. The number who catch it is recorded.	**j**	Five cards are drawn one at a time, **without replacement**, from a shuffled deck. The number of hearts is recorded.

Yes: a, c, e, f

Binomial Random Variables

A binomial random variable is defined by:

number of trials

$$X \sim \text{Bin}(n, p)$$

probability of 'success' for each trial

Example Write $X \sim \text{Bin}(n, p)$ for a binomial distribution

A 20-question multiple-choice quiz is set. Each question has **four** options, one of which is correct. A student guesses the answer to every question at random, and her answer to one question has no effect on her answer to any other. The number of questions she answers correctly is recorded.

$X \sim \text{Bin}(20, 0.25)$

Guided Practice

a Same quiz, same student as above, but each question has **five** options.

$X \sim \text{Bin}(20, 0.2)$

b Same five-option quiz, but there are only **12 questions**.

$X \sim \text{Bin}(12, 0.2)$

c Same five-option quiz of 20 questions but the student's score is recorded as **the number she gets wrong**.

$X \sim \text{Bin}(20, 0.8)$

d A machine fills 500 mL bottles. 4% of bottles are underfilled. An inspector takes 30 bottles at random from a pallet of 6000 and records how many are underfilled

$X \sim \text{Bin}(30, 0.04)$

Key Ideas

1. A random variable X is binomial if it satisfies these conditions:
   a. the number of trials n is ……
   b. each trial has exactly …… outcomes
   c. the trials are ……
   d. the probability of success p is …… in every trial
2. X counts the number of ……, we write $X \sim \text{Bin}(n, p)$

1 State whether each random variable describes a binomial distribution.

a The number of heads when tossing a coin 20 times

Yes: fixed number of independent trials with constant $p = \frac{1}{2}$.

b The number of hits when throwing a ball at a target 45 times

Yes: fixed number of trials with hit or miss outcomes.

c The number of white marbles when 10 marbles are randomly selected from a bag of coloured marbles without replacement

No: without replacement, the probability changes.

d The number of defective batteries in a sample of 100 batteries

Yes: defective or not defective over a fixed sample.

e The total of 2 dice when they are rolled 50 times

No: the total of two dice has more than 2 possible outcomes per trial.

2 For each binomial context, define success, define X in words, state n and p, and write $X \sim \text{Bin}(n, p)$.

a A fair die is rolled 45 times.
The number of times it shows a number greater than 4 is recorded.

Success = the die shows 5 or 6. X = number of rolls out of 45 showing more than 4. $X \sim \text{Bin}(45, 1/3)$.

b Seven per cent of a large batch of seeds fail to germinate. Forty seeds are planted.
The number that germinate is recorded.

Success = the seed germinates. X = number of the 40 seeds that germinate. $X \sim \text{Bin}(40, 0.93)$.

c A student guesses all 25 questions of a true/false quiz.
The number answered incorrectly is recorded.

Success = the question is answered incorrectly. X = number of the 25 questions answered incorrectly. $X \sim \text{Bin}(25, 0.5)$.

d A card is drawn from a shuffled deck, recorded, and replaced. This is repeated 18 times.
The number of picture cards (J, Q, K) drawn is recorded.

Success = the card is a picture card. X = number of picture cards in 18 draws. $X \sim \text{Bin}(18, 12/52) = \text{Bin}(18, 3/13)$.

e In a large city, 62% of adults hold a driver's licence. Fifty adults are chosen at random.
The number without a licence is recorded.

Success = the adult does not hold a licence. X = number of the 50 adults without a licence. $X \sim \text{Bin}(50, 0.38)$.

f A basketballer with a free-throw rate of 0.85 takes 16 shots, resting fully between each.
The number she misses is recorded.

Success = the shot is missed. X = number of the 16 shots missed. $X \sim \text{Bin}(16, 0.15)$.

Probability of r successes

Review

☐ Calculate multistage probability of independent events

A basketball player scores a three-point shot with probability 0.3.
Each shot is independent.

a She takes two shots. Find the probability she scores both.

b She takes two shots. Find the probability she scores the first and misses the second.

c She takes three shots. Find the probability she misses all three.

d She takes four shots. Write, as a product of powers, the probability she scores the first two and misses the last two.

☐ Count arrangements of successes and failures

a How many arrangements of two S's and two F's are there in four trials?

b How many arrangements of two Ss and four Fs are there in six trials?

☐ State 'success' event, then n then p of a binomial distribution

a A player takes 6 three-point shots. Each is scored with probability 0.3.

Success: ……
n: ……
p: ……

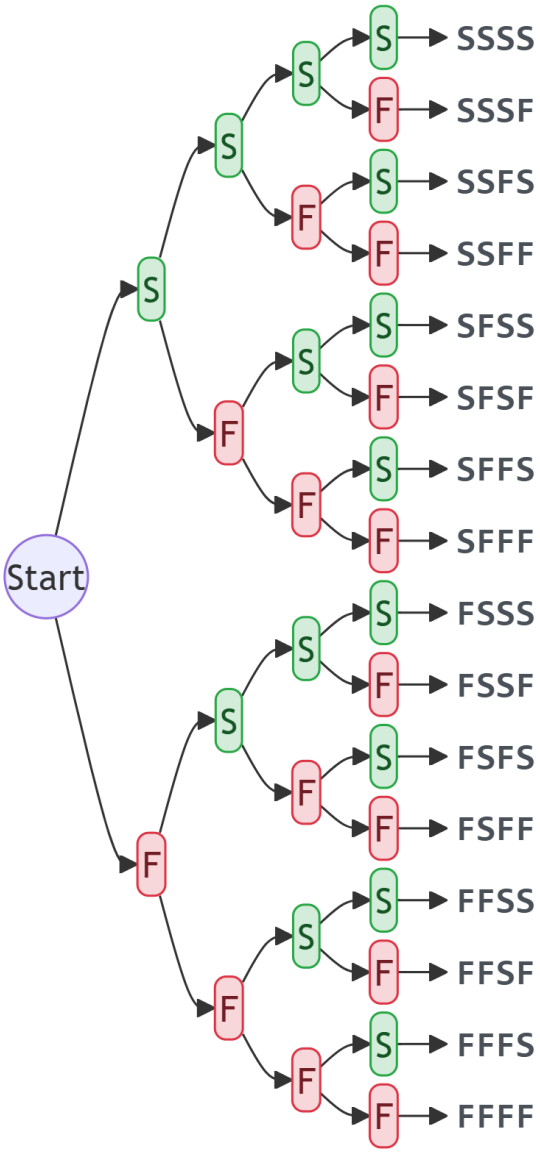
b A die is rolled 10 times. We are interested in the number of sixes.

c Twelve components are tested. Each is defective with probability 0.1.
We are interested in the number of defectives.

Investigation binomial distributions

A fair die is rolled 4 times and we record the number of times six is rolled.

We will take *six* to represent a 'success' and *not six* to be a 'failure'.



a What is the probability of four successes in a row S S S S?

b What is the probability of four failures in a row F F F F?

c What is the probability of S S S F, in that order?

d What is the probability of S S F S, in that order?

e What is the probability of 3 S, 1 F in any order?

f What is the probability of S S F F, in that order?

g How many ways are there to get 2 S and 2 F?

h Therefore, what is the probability of 2 S and 2 F, in any order?

Suppose an experiment is repeated several times, and each trial results in either a success or a failure.

To find the probability of one specific sequence of successes and failures, such as S S S F, we ..... the probabilities of the individual outcomes.

Since we can multiply in any order without changing the result, two sequences have the same probability when they contain the same number of successes and failures, so S S S F and ..... have the same probability.

To find the probability of exactly 3 successes out of 4 in any order, we first count the number of possible ..... containing 3 successes out of 4. We then multiply the number of arrangements by the probability of each such arrangement.

ⓘ Probability of r successes

For a binomial random variable $X \sim \text{Bin}(n, p)$, the probability of r successes is:

number of ways to arrange r successes out of n

$$P(X = r) = {}^{n}C_r \, p^r (1-p)^{n-r}$$

probability of each sequence of r successes and $n - r$ failures

ⓘ Interpret probability of r successes formula

$X \sim \text{Bin}(6,0.3)$, find $P(X = 2)$.	$X \sim \text{Bin}(6,0.3)$, find $P(X = 3)$.
$P(X = 2) = {}^{6}C_2 \cdot 0.3^2 \cdot (1-0.3)^4$	$P(X = 3) = {}^{6}C_3 \cdot 0.3^3 \cdot (1-0.3)^3$
Find the probability of:	
a $X \sim \text{Bin}(6, 0.4)$, $P(X = 1)$	**b** $X \sim \text{Bin}(6, 0.3)$, $P(X = 2)$
c $X \sim \text{Bin}(8, 0.3)$, $P(X = 2)$	**d** $X \sim \text{Bin}(7, 0.4)$, $P(X = 2)$
e $X \sim \text{Bin}(9, 0.15)$, $P(X = 3)$	**f** $X \sim \text{Bin}(5, 0.2)$, $P(X = 2)$

Example calculate probability of r successes in a binomial distribution

A basketball player scores a shot with probability 0.3.

She takes 6 shots, with each shot independent of the previous.

Let X be the number of shots that she scores. Find the probability she gets 2 shots in.

$X \sim \text{Bin}(6,0.3)$

$n = 6,\ p = 0.3,\ r = 2$

$P(X = 2) = {}^{6}C_{2} \cdot 0.3^{2} \cdot (1 - 0.3)^{4}$

$= 0.324135$

Guided Practice

a Same player as above, same 6 shots, same probability 0.3.
Find the probability she gets 4 shots in.

0.0595

b The player improves. She now scores with probability 0.35. She still takes 6 shots.
Find the probability she gets 4 shots in.

0.0951

c She takes 7 shots instead of 6, still scoring with probability 0.35. Find $P(X = 4)$.

0.1442

d A machine produces components. Each component is defective with probability 0.15, independently of the others. A sample of 8 components is taken. Find the probability that exactly 3 are defective, correct to four decimal places.

0.0839

Key Ideas

1. If $X \sim \text{Bin}(n,p)$, then $P(X = r) =$ ____ $\times p^{-} \times (1 - p)^{--}$.
2. The factor ${}^{n}C_{r}$ counts the number of …… in which r successes can occur in n trials.
3. The exponent r is the number of …… and the exponent $n - r$ is the number of ……
4. The two exponents always add to ……
5. Every arrangement with r successes has the …… probability, which is why the arrangements may be counted and then multiplied.

ⓘ Binomial Distribution on a Calculator

Casio FX 8200AU:

1. Press HOME ⌂

2. Select Distribution

3. Select [Binomial PD] (binomial probability distribution)

To see the binomial probability distribution of $X \sim \text{Bin}(6, 0.3)$:

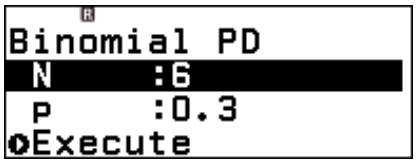
a. [List]

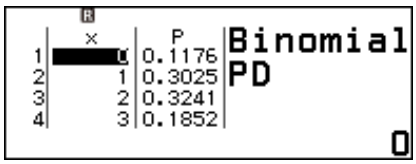
b. Enter data from 0 to 6. Here, x represents number of successes, r.
(if you want a specific number of successes, just enter that one).

c. Press OK and enter the number of trials N and probability of success p

d. Press ▶Execute







You can see that $P(X = 2) = 0.3241$ like we calculated using ${}^6C_2 \cdot 0.3^2 \cdot (1 - 0.3)^4$

1 $X \sim \text{Bin}(5,0.2)$. Find $P(X = 2)$.

${}^5C_2 \times 0.2^2 \times 0.8^3 = 0.2048$

2 $X \sim \text{Bin}(5,0.2)$. Find $P(X = 3)$

${}^5C_3 \times 0.2^3 \times 0.8^2 = 0.0512$

3 $X \sim \text{Bin}(7,0.4)$. Find $P(X = 2)$

${}^7C_2 \times 0.4^2 \times 0.6^5 \approx 0.2613$

4 $X \sim \text{Bin}(9,0.6)$. Find $P(X = 5)$

${}^9C_5 \times 0.6^5 \times 0.4^4 \approx 0.2508$

5 A fair die is rolled 6 times. Find the probability that exactly 2 of the rolls show a three.

Success: the roll shows a three.

${}^6C_2 \times \left(\frac{1}{6}\right)^2 \times \left(\frac{5}{6}\right)^4 \approx 0.2009$

6 A player scores each three-point shot with probability 0.3, independently. She takes 8 shots. Find the probability she scores exactly 3.

${}^8C_3 \times 0.3^3 \times 0.7^5 \approx 0.2541$

7 In a one-day cricket game, a batsman has a chance of $\frac{1}{5}$ of hitting a boundary every time he faces a ball. If he faces all six balls in an over, what is the probability that he will hit exactly two boundaries, assuming that successive strikes are independent?

Let $X \sim B(6,\frac{1}{5})$.

$P(X = 2) = \binom{6}{2}\left(\frac{1}{5}\right)^2\left(\frac{4}{5}\right)^4 = 0.24576$

8 If the probability of a manufacturer's machine breaking down is $\frac{3}{11}$, find the probability that at any particular time 7 machines out of a total of 20 machines will be broken down. Leave your answer in index form.

$X \sim \text{Bin}\left(20,\frac{3}{11}\right)$

$P(X = 7) = \binom{20}{7}\left(\frac{3}{11}\right)^7\left(\frac{8}{11}\right)^{13}$

9 The probability that a small earthquake occurs somewhere in the world on any one day is 0.95. Assuming that earthquake frequencies on successive days are independent, what is the probability that a small earthquake occurs somewhere in the world on exactly 28 of January's 31 days? Leave your answer in index form.

Let $X \sim B(31,0.95)$.

$P(X = 28) = \binom{31}{28}(0.95)^{28}(0.05)^3$

10 A survey showed that about 70% of people eat Crunchy Muesli for breakfast. If another survey is carried out with 20 people, find the probability that:

a 12 people eat Crunchy Muesli

b 13 people eat Crunchy Muesli

c 9 people eat Crunchy Muesli

$X \sim \text{Bin}(20,0.7)$

$P(X = 12) = \binom{20}{12}(0.7)^{12}(0.3)^{8} = 11.4\%$

$P(X = 13) = \binom{20}{13}(0.7)^{13}(0.3)^{7} = 16.4\%$

$P(X = 9) = \binom{20}{9}(0.7)^{9}(0.3)^{11} = 1.2\%$

11 A quiz has 20 multiple-choice questions, each with a choice of 4 answers. If Jay guesses an answer for each question at random, find the probability that he will score 75% on this quiz, correct to 2 significant figures.

75% of 20 questions is 15 correct answers.

Let $X \sim \text{Bin}\left(20, \frac{1}{4}\right)$

$P(X = 15) = \binom{20}{15}\left(\frac{1}{4}\right)^{15}\left(\frac{3}{4}\right)^{5} = 3.4 \times 10^{-6}$

12 A jury roll contains 2000 names, 700 of females and 1300 of males. Twelve jurors are randomly selected.

a Explain why it is reasonable to make the approximation that the probability of selecting a male does not change with each selection.

b What is the probability of ending up with an all-male jury?

Only 12 of 2000 names are removed, so the male proportion changes very little. Use $p = \frac{1300}{2000} = 0.65$ for each selection.

$P(\text{all male}) = (0.65)^{12} = 0.005688$

13 A casino has 20 poker machines and 8 roulette wheels. The probability of a poker machine being faulty is $\frac{1}{6}$ and the probability of a roulette wheel being faulty is $\frac{3}{11}$.
Find the probability that at a certain time 3 poker machines and 2 roulette wheels are faulty. Leave your answer in index form.

$P = \binom{20}{3}\left(\frac{1}{6}\right)^{3}\left(\frac{5}{6}\right)^{17}\binom{8}{2}\left(\frac{3}{11}\right)^{2}\left(\frac{8}{11}\right)^{6}$

14 A man is restoring ten old cars, six of them manufactured in 1955 and four of them manufactured in 1962. When he tries to start them, on average the 1955 models will start 65% of the time and the 1962 models will start 80% of the time. Find, correct to four decimal places, the probability that at any time:

a exactly three of the 1955 models and one of the 1962 models will start,

b exactly four of the cars will start. Consider five cases.

Let $X \sim B(6,0.65)$ for the 1955 cars and $Y \sim B(4,0.8)$ for the 1962 cars.

$$P(X = 3, Y = 1) = \binom{6}{3}(0.65)^3(0.35)^3\binom{4}{1}(0.8)(0.2)^3 = 0.0060$$

Exactly four cars start when (X, Y) is $(0,4)$, $(1,3)$, $(2,2)$, $(3,1)$ or $(4,0)$.

$$P(X + Y = 4) = \sum_{x=0}^{4}\binom{6}{x}(0.65)^x(0.35)^{6-x}\binom{4}{4-x}(0.8)^{4-x}(0.2)^x = 0.0303$$

15 $X \sim \text{Bin}(6,0.3)$

a Explain why ${}^6C_2 = {}^6C_4$ without evaluating.

Choosing which 2 of the 6 trials are successes is the same as choosing which 4 are failures, so the two counts are equal

b Find $P(X = 2)$ and $P(X = 4)$.

$P(X = 2) \approx 0.3241$; $P(X = 4) \approx 0.0595$

c The counting factors are equal. Explain why the two probabilities are not.

The counting factor is 15 in both, but the bases carry different exponents: $0.3^2 \times 0.7^4 = 0.021609$ against $0.3^4 \times 0.7^2 = 0.003969$.

The probabilities are only equal if it was an equal probability of success and failure.

16 $X \sim \text{Bin}(5, p)$ and $P(X = 5) = 0.07776$. Find p.

$${}^5C_5 \times p^5 \times (1-p)^0 = p^5$$

$$p^5 = 0.07776$$

$$p = 0.6.$$

17 A student writes, for $X \sim \text{Bin}(9,0.2)$: $P(X = 4) = {}^9C_4 \times 0.2^4 \times 0.8^4$. Identify the error, state the check that would have caught it, and give the correct value.

The failure exponent should be $9 - 4 = 5$, not 4. The check is that the exponents must add to $n = 9$.

Correct value: ${}^9C_4 \times 0.2^4 \times 0.8^5 \approx 0.0661$

At Least, At Most, More Than

Review

☐ Recall language of compound events

A set of numbers is $\{0, 1, 2, 3, 4, 5, 6\}$

Write the subset for:

a	More than 2	**b**	At least 2
c	2 or more	**d**	Not fewer than 2
e	Fewer than 2	**f**	At most 2
g	Not more than 2	**h**	Not more than 5
i	Between 2 and 4, inclusive	**j**	No more than 2

A binomial distribution is $X \sim \text{Bin}(n, p)$ write the set of r for these phrases:

a	$X \sim \text{Bin}(5,0.3)$ At least 2	**b**	$X \sim \text{Bin}(4,0.3)$ At least 2	**c**	$X \sim \text{Bin}(4,0.3)$ No more than 2

☐ Identify complement set

Sometimes it is more efficient to do $P(A) = 1 - P(\bar{A})$ if the complement set is smaller.

For $X = \{0,1,2,3,4,5,6\}$, identify the direct set and the complement set and decide whether finding the complement would be more efficient.

a	$X \geq 4$	**b**	$X \geq 2$
c	$X \leq 5$	**d**	$X > 1$
e	At least 2	**f**	At most 3

Example Solve binomial distribution problems involving at most, at least etc.

A machine produces components. 30% of components are faulty.

A sample of 6 components is taken.

Find the probability that at least 2 components are faulty.

Success: a component is faulty

$n = 6,\ p = 0.3,$

$X \sim \text{Bin}(6,0.3)$

"At least 2" means $r \in \{2,3,4,5,6\}$; 5 terms

Complement $\{0,1\}$: 2 terms.

$P(X \geq 2) = 1 - P(X = 0) - P(X = 1)$

$P(X = 0) = {}^6C_0(0.3)^0(1-0.3)^6 = (0.7)^6$

$P(X = 1) = {}^6C_1(0.3)^1(1-0.3)^5 = 6(0.3)(0.7)^5$

$\therefore P(X \geq 2) = 1 - (0.7)^6 - 6(0.3)(0.7)^5$

≈ 0.5798

Guided Practice

a $X \sim \text{Bin}(6,0.3)$. Find $P(X > 2)$.

0.2557

b $X \sim \text{Bin}(6,0.3)$. Find $P(X \leq 2)$.

0.7443

c A set of traffic lights is red on 20% of approaches. A driver passes 5 such lights. Find the probability that **at most 1** is red.

0.7373

d $X \sim \text{Bin}(6,0.3)$. Find $P(2 \leq X \leq 4)$.

0.5689

ⓘ Binomial Cumulative Probabilities using a Calculator

Casio FX 8200AU:

1. Press HOME ⌂

2. Select Distribution

3. Select [Binomial CD] (binomial cumulative distribution; this finds $P(X \leq x)$)

To see the cumulative binomial probability distribution of $X \sim \text{Bin}(6,0.3)$:

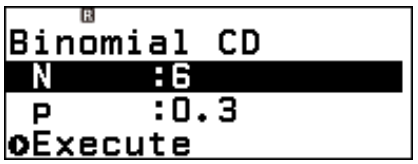
a. [List]

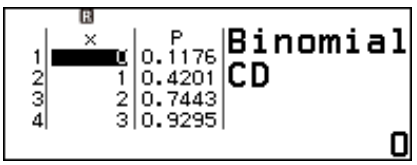
b. Enter data from 0 to 6 (if you want a specific number of successes, just enter that one).

c. Press OK and enter the number of trials N and probability of success p

d. Press ●Execute





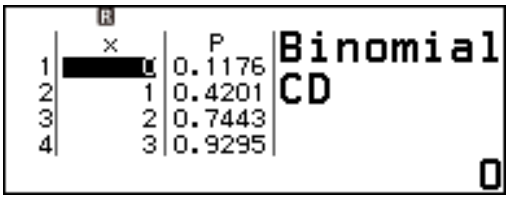


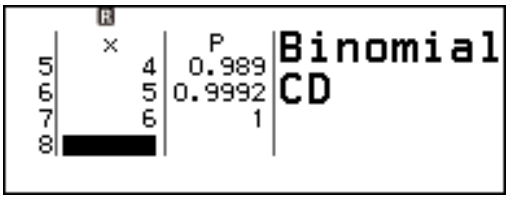
The values show $P(X \leq x)$.

To find $P(X \geq 2)$, you would do $1 - P(X \leq 1) = 1 - 0.4201 = 0.5798$, as we found earlier.

ⓘ **Interpret** Binomial cumulative distribution

Binomial CD for $X \sim \text{Bin}(6,0.3)$ is shown.





Find:

a	$P(X \leq 3)$	**b**	$P(X < 3)$	**c**	$P(X > 3)$
	0.9295		0.7443		0.0705

1 A die is thrown 4 times. Find the probability of throwing:

a 2 threes

$$P(X=2)=\binom{4}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^2=\frac{25}{216}$$

b 3 ones

$$P(X=3)=\binom{4}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)=\frac{5}{324}$$

c 1 four

$$P(X=1)=\binom{4}{1}\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^3=\frac{125}{324}$$

d fewer than 3 ones

$$P(X<3)=1-P(X=3)-P(X=4)=\frac{425}{432}$$

e at least 3 fives

$$P(X\geq 3)=P(X=3)+P(X=4)=\frac{7}{432}$$

2 The probability that dogs of a certain breed will have black spots is $\frac{2}{5}$.
If a dog of this breed has 5 pups, find the probability that:

a 3 will have black spots

$$P(X=3)=\binom{5}{3}\left(\frac{2}{5}\right)^3\left(\frac{3}{5}\right)^2=\frac{144}{625}$$

b none will have black spots

$$P(X=0)=\left(\frac{3}{5}\right)^5=\frac{243}{3125}$$

c 4 will have black spots

$$P(X=4)=\binom{5}{4}\left(\frac{2}{5}\right)^4\left(\frac{3}{5}\right)=\frac{48}{625}$$

d at least 3 will have black spots

$$P(X\geq 3)=\mathrm{P(X=3)+P(X=4)+P(X=5)}=\frac{992}{3125}$$

e fewer than 3 will have black spots

$$P(X<3)=1-P(X\geq 3)=\frac{2133}{3125}$$

3 A die is rolled six times. Let N denote the number of times that the number 3 is shown on the uppermost face. Find, correct to four decimal places:

a $P(N=2)$

b $P(N<2)$

c $P(N\geq 2)$

Let $N\sim B(6,\frac{1}{6})$.

$$P(N=2)=\binom{6}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^4=0.2009$$

$$P(N<2)=\left(\frac{5}{6}\right)^6+6\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^5=0.7368$$

$$P(N\geq 2)=1-P(N<2)=0.2632$$

4 A poll indicates that 55% of people support the policies of the Working Together Party. If five people are selected at random, what is the probability that a majority of them will support WTP party policies? Give your answer correct to three decimal places.

Let $X \sim B(5,0.55)$.

$$P(X \geq 3) = \sum_{x=3}^{5} \binom{5}{x} (0.55)^x (0.45)^{5-x} = 0.593$$

5 The probability of an arrow hitting a target is 0.7. If 8 arrows are fired, find the probability of hitting the target:

a 5 times

$$P(X = 5) = \binom{8}{5} (0.7)^5 (0.3)^3 = 0.254$$

b Twice

$$P(X = 2) = \binom{8}{2} (0.7)^2 (0.3)^6 = 0.0100$$

c 3 times

$$P(X = 3) = \binom{8}{3} (0.7)^3 (0.3)^5 = 0.0467$$

d 7 times

$$P(X = 7) = \binom{8}{7} (0.7)^7 (0.3) = 0.198$$

e at least 6 times

$$P(X \geq 6) = \mathrm{P(X = 6) + P(X = 7) + P(X = 8)} = 0.552$$

6 An archer finds that on average he hits the bulls-eye nine times out of ten. Assuming that successive attempts are independent, find the probability that in twenty attempts:

a he scores at least eighteen hits,

b he misses at least once.

Let $X \sim B(20,0.9)$.

$$P(X \geq 18) = \binom{20}{18} (0.9)^{18} (0.1)^2 + \binom{20}{19} (0.9)^{19} (0.1) + (0.9)^{20} = 0.67693$$

$$P(\text{at least one miss}) = 1 - (0.9)^{20} = 0.87842$$

7 A torch manufacturer finds that on average 9% of the bulbs are defective. What is the probability that in a randomly selected batch of ten one-bulb torches:

a there will be no more than two with defective bulbs,

b there will be at least two with defective bulbs?

Let $X \sim B(10,0.09)$.

$$P(X \leq 2) = (0.91)^{10} + \binom{10}{1} (0.09)(0.91)^9 + \binom{10}{2} (0.09)^2 (0.91)^8 = 0.94596$$

$$P(X \geq 2) = 1 - (0.91)^{10} - \binom{10}{1} (0.09)(0.91)^9 = 0.22545$$

8 Kim throws 8 dice. Find the probability that she throws:

a 2 sixes

$$P(X=2)=\binom{8}{2}\left(\frac{1}{6}\right)^2\left(\frac{5}{6}\right)^6=\frac{109375}{419904}$$

b 5 sixes

$$P(X=5)=\binom{8}{5}\left(\frac{1}{6}\right)^5\left(\frac{5}{6}\right)^3=\frac{875}{209952}$$

c 6 sixes

$$P(X=6)=\binom{8}{6}\left(\frac{1}{6}\right)^6\left(\frac{5}{6}\right)^2=\frac{175}{419904}$$

d at least 7 sixes

$$P(X\geq 7)=P(X=7)+P(X=8)=\frac{41}{1679616}$$

e fewer than 2 sixes

$$P(X<2)=P(X=0)+P(X=1)=\frac{1015625}{1679616}$$

9 A die is rolled twelve times. Find the probability that 5 appears on the uppermost face:

a exactly three times,

b exactly eight times,

c ten or more times, that is, ten, eleven or twelve times.

Let $X\sim B(12,\frac{1}{6})$.

$$P(X=3)=\binom{12}{3}\left(\frac{1}{6}\right)^3\left(\frac{5}{6}\right)^9$$

$$P(X=8)=\binom{12}{8}\left(\frac{1}{6}\right)^8\left(\frac{5}{6}\right)^4$$

$$P(X\geq 10)=\binom{12}{10}\left(\frac{1}{6}\right)^{10}\left(\frac{5}{6}\right)^2+\binom{12}{11}\left(\frac{1}{6}\right)^{11}\left(\frac{5}{6}\right)+\left(\frac{1}{6}\right)^{12}$$

10 A coin is tossed 10 times, and a student is asked to calculate the probability that more than 75% of the coins will show heads.

a If more than 75% of the coins show heads, how many coins would this be?

b Use the binomial probability formula to determine the probability of obtaining more than 75% heads. On this occasion the calculation is easily done without a normal approximation.

More than 75% of 10 means 8, 9 or 10 heads.

Let $X\sim B(10,0.5)$.

$$P(X\geq 8)=\frac{\binom{10}{8}+\binom{10}{9}+\binom{10}{10}}{2^{10}}=0.0547=5.47\%$$

11 During winter it rains on average 18 out of 30 days. Five winter days are selected at random. Find, correct to four decimal places, the probability that:

a the first two days chosen will be fine and the remainder wet,

b more rainy days than fine days have been chosen.

Use $P(\text{wet}) = \frac{18}{30} = 0.6$ and $P(\text{fine}) = 0.4$.

$P(\text{first two fine, next three wet}) = (0.4)^2(0.6)^3 = 0.03456$

Let $X \sim B(5,0.6)$ record rainy days.

$$P(X \geq 3) = \sum_{x=3}^{5} \binom{5}{x} (0.6)^x (0.4)^{5-x} = 0.68256$$

12 The probability that a jackpot prize will be won in a given lottery is 0.012.

a Find, correct to five decimal places, the probability that the jackpot prize will be won:

i exactly once in ten independent lottery draws,

ii at least once in ten independent lottery draws.

Let $X \sim B(10,0.012)$.

$$P(X = 1) = \binom{10}{1} (0.012)(0.988)^9 = 0.10764$$

$$P(X \geq 1) = 1 - (0.988)^{10} = 0.11372$$

b The jackpot prize is initially $\$10\,000$ and increases by $\$10\,000$ each time the prize is not won. Find, correct to five decimal places, the probability that the jackpot prize will exceed $\$200\,000$ when it is finally won.

The prize first exceeds $\$200\,000$ after 20 consecutive draws without a win.

$P(\text{prize exceeds } \$200\,000) = (0.988)^{20} = 0.78549$

13 Five families have three children each.

a Find, correct to three decimal places, the probability that:

b at least one of these families has three boys,

c each family has more boys than girls.

d What assumptions have been made in arriving at your answer?

For one family, $P(\text{three boys}) = \frac{1}{8}$.

$$P(\text{at least one}) = 1 - \left(\frac{7}{8}\right)^5 = 0.487$$

For one family, $P(\text{more boys than girls}) = \frac{4}{8} = \frac{1}{2}$.

$$P(\text{all five families}) = \left(\frac{1}{2}\right)^5 = 0.031$$

Assume boys and girls are equally likely, births are independent, and every child is classified as a boy or a girl.

14 In a game, a person wins if a die comes up with the number 4 when rolled.

a If X is the discrete random variable for getting a 4 when rolling a die 6 times, find all the possible values of X.

b Draw up a probability distribution table to show all the probabilities, as decimals, of getting a 4 when rolling the die 6 times.

c Draw a bar graph to show this information.

d Describe the shape of the graph.

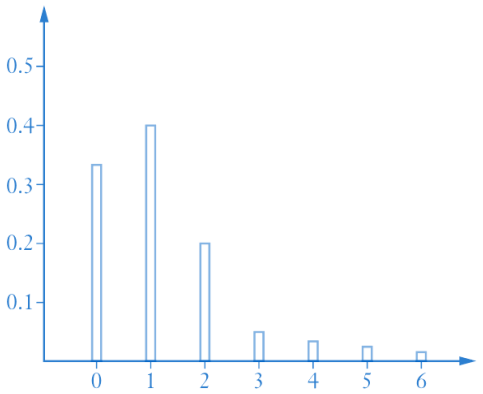
e Find the probability that a 4 will come up at least 3 times if the die is rolled 6 times.

$X \in \{0,1,2,3,4,5,6\}$

$X \sim \text{Bin}(6,\frac{1}{6})$.

$P(X = x) = \binom{6}{x}\left(\frac{1}{6}\right)^x\left(\frac{5}{6}\right)^{6-x}$

x	0	1	2	3	4	5	6
$P(X = x)$	0.33	0.40	0.20	0.05	0.008	0.0006	0.0000



The graph is positively skewed.

Using unrounded probabilities:

$P(X \geq 3) = \sum_{x=3}^{6}\binom{6}{x}\left(\frac{1}{6}\right)^x\left(\frac{5}{6}\right)^{6-x} = 0.06229$

15 One bag contains three red and five white balls, and another bag contains four red and four white balls.

a One bag is chosen at random, a ball is selected from that bag, its colour is noted, and then it is replaced. Find the probability that the ball chosen is red.

b If the operation in part a is carried out eight times, find the probability that:

c exactly three red balls are drawn,

d at least three red balls are drawn.

$P(\text{red}) = \frac{1}{2}\left(\frac{3}{8}\right) + \frac{1}{2}\left(\frac{4}{8}\right) = \frac{7}{16}$

Let $X \sim B(8,\frac{7}{16})$.

$P(X = 3) = \binom{8}{3}\left(\frac{7}{16}\right)^3\left(\frac{9}{16}\right)^5$

$P(X \geq 3) = 1 - \left(\frac{9}{16}\right)^8 - 8\left(\frac{7}{16}\right)\left(\frac{9}{16}\right)^7 - \binom{8}{2}\left(\frac{7}{16}\right)^2\left(\frac{9}{16}\right)^6$

16 An apple exporter deals in two types of apples, Red Delicious and Golden Delicious. The ratio of Red Delicious to Golden Delicious is $4:1$. The apples are randomly mixed together before they are boxed. One in every fifty Golden Delicious and one in every one hundred Red Delicious apples will need to be discarded because they are undersized.

a What is the probability that an apple selected from a box will need to be discarded?

b If ten apples are randomly selected from a box, find the probability that:

i all the apples will have to be discarded,

ii half of the apples will be discarded,

iii fewer than two apples will be discarded.

$$P(\text{discard}) = \frac{4}{5}\left(\frac{1}{100}\right) + \frac{1}{5}\left(\frac{1}{50}\right) = \frac{3}{250}$$

Let $X \sim B(10, \frac{3}{250})$.

$$P(X = 10) = \left(\frac{3}{250}\right)^{10}$$

$$P(X = 5) = \binom{10}{5}\left(\frac{3}{250}\right)^{5}\left(\frac{247}{250}\right)^{5}$$

$$P(X < 2) = \left(\frac{247}{250}\right)^{10} + 10\left(\frac{3}{250}\right)\left(\frac{247}{250}\right)^{9}$$

17 A tennis player finds that, on average, he gets his serve in eight out of every ten attempts and that he serves an ace, a serve which is in the boundaries and not touched by his opponent, once every fifteen serves. He serves four times. Assuming that successive serves are independent events, find, correct to six decimal places, the probability that:

a all four serves are in,

b he hits at least three aces,

c he hits exactly three aces and the other serve lands in.

$$P(\text{all four in}) = (0.8)^4 = 0.409600$$

Let $X \sim B(4, \frac{1}{15})$ record aces.

$$P(X \geq 3) = \binom{4}{3}\left(\frac{1}{15}\right)^{3}\left(\frac{14}{15}\right) + \left(\frac{1}{15}\right)^{4} = 0.001126$$

$$P(\text{in but not an ace}) = \frac{4}{5} - \frac{1}{15} = \frac{11}{15}.$$

$$P = \binom{4}{3}\left(\frac{1}{15}\right)^{3}\left(\frac{11}{15}\right) = 0.000869$$

18 How many times must a die be rolled so that the probability of rolling at least one six is greater than 95%?

$$1 - \left(\frac{5}{6}\right)^{n} > 0.95$$

$$n > \frac{\log(0.05)}{\log(5/6)}$$

Minimum $n = 17$.

Mean, Variance, Expected Frequency of Binomial Distributions

Investigation Mean, variance of binomial distributions

Suppose a basketball player is taking a free throw, and he has an 80% chance of getting it in.

X is a …… random variable, with success probability …… and failure probability ……

x	0	1
$P(x)$		

The expected value is ……

The variance is ……

Suppose now that the same player is taking $n = 10$ independent free throws, each with the same success probability of 80%.

This is a …… random variable. You would *expect* him to score …… shots.

This is due to the **linearity of expectation** (not strictly in the course)

$$E(X + Y) = E(X) + E(Y) \text{ and } E(aX + b) = aE(X) + b$$

Variance add for independent random variables.

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

The expected value for a Bernoulli random variable is p and variance is $p(1 - p)$;

so, for a binomial random variable, expected value is …… and variance is ……

Variance is not linear

Be very careful with variance.

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$$

The above relationship is only true for independent events.

Variance is not linear:

$$\text{Var}(aX) = a^2 \text{ Var}(X)$$
$$\text{Var}(3X) = 9 \text{ Var}(X)$$

Imagine X is a random jump of either $+1$ or -1 metres.

For $X + Y$, you could jump forward 1 and jump backwards 1, the spread stays controlled as they can cancel out.

For $X + X + X$, this is a single jump tripled (+1 metre would become +3 metres), so the final spread is much wider as you don't have randomness cancelling each other out.

ⓘ Mean, Variance, Standard Deviation of Binomial Distributions

Let a binomial random variable X be n repetitions of a Bernoulli trial; we multiply the Bernoulli expected value and variance by n:

Expected Value	Variance & Standard Deviation
$E(X) = np$	$\text{Var}(X) = np(1-p)$ $\sigma(X) = \sqrt{np(1-p)}$

⚠ $\sigma(X)$ is **not** n times the Bernoulli standard deviation.
$\sigma(X) \neq n\sqrt{p(1-p)}$ as standard deviation does not have the linearity property.

ⓘ n or N

When solving problems related to expected frequency, we need to read the question carefully.
n is the number of trials **inside one experiment**.
N is the number of times the **whole experiment** is repeated.

ⓘ Interpret each number in a binomial probability scenario

A batch contains 40 items. Each item is defective with probability 0.15.
200 such batches are inspected. How many batches contain exactly 2 defective items?

40 = n 0.15 = p 200 = N 2 = r

Identify whether each number represents n, p, N or r

a In 180 packets of 12 seeds, how many packets have exactly 3 germinate, if each seed has an 80% chance of germinating?

b 120 boxes each hold 25 eggs. Each egg is cracked with probability 0.04. In how many boxes are exactly 3 eggs cracked?

c A quiz has 20 multiple-choice questions. 300 students guess every answer, each with probability 0.25 of being right. How many students get exactly 5 correct?

d A batch of 40 items has each item defective with probability 0.15. Find the probability that exactly 2 are defective.

Expected Frequency

For entire experiment repeated N times, the expected number of repetitions in which $X = r$ is:

$$N \cdot P(X = r)$$

Example Calculate mean, sd, and expected frequency of a binomial distribution

A batch contains 40 items. Each item is defective, independently, with probability 0.15.
Let X be the number of defective items in a batch.

a Find $E(X)$, $\text{Var}(X)$ and $\text{sd}(X)$.
b 200 such batches are inspected.
Find the expected number of batches containing exactly 2 defective items.

a.

$E(X) = np = 40(0.15) = 6$

$\text{Var}(X) = np(1-p) = 40(0.15)(0.85) = 5.1$

$\sigma(X) = \sqrt{np(1-p)} = \sqrt{5.1} \approx 2.26$ (2 d.p.)

b.

$N \cdot P(X = 2) = 200 \cdot \binom{40}{2} (0.15)^2 (0.85)^{38}$

≈ 7 batches

Note that being defective is taken to be 'success' here

Guided Practice

a A batch contains 40 items. Each item is defective with probability **0.2**.
Find $E(X)$ and $\text{Var}(X)$.

$E(X) = 8, \text{Var}(X) = 6.4$

b A batch of 30 items has each item defective with probability 0.1. In 500 batches, how many are expected to contain exactly 3 defectives?

118

c $X \sim \text{Bin}(40, 0.15)$. Of 200 batches, how many are expected to contain **at least 2** successes?

$200 \times P(X \geq 2) = 198$ batches

d $X \sim \text{Bin}(n, p)$ with $E(X) = 12$ and $\text{Var}(X) = 9.6$. Find n and p.

$p = 0.2, n = 60$

1 The mean of a binomial distribution is 20.5.
If there are 50 trials, find the probability of success as a fraction.

Using $np = 20.5$:

$$p = \frac{20.5}{50} = \frac{41}{100}$$

2 The mean of a binomial distribution is 12.6 and the probability of success is $\frac{1}{5}$.
Find the number of trials.

Using $np = 12.6$:

$$n = \frac{12.6}{1/5} = 63$$

3 The mean of a binomial distribution is 12.92 and the probability of failure is 0.32.
Find the number of trials.

$p = 1 - 0.32 = 0.68.$

$$n = \frac{12.92}{0.68} = 19$$

4 Find the binomial distribution with 50 trials that has mean 30.

Using $np = 30$ with $n = 50$:

$$p = \frac{30}{50} = \frac{3}{5}$$

Therefore $X \sim \text{Bin}\left(50, \frac{3}{5}\right)$

5 The probability of a new car passing a quality-control check is 93%. If a quality-control check is made on 250 cars, then for the number of cars that pass, find:

a the mean
b the variance
c the standard deviation

$X \sim \text{Bin}(250, 0.93)$

$E(X) = 250(0.93) = 232.5$

$\text{Var}(X) = 250(0.93)(0.07) = 16.275$

$\sigma = \sqrt{16.275} = 4.034$

6 The probability of selecting a black jelly bean from a packet is 0.16.

a Find the mean number of black jelly beans selected from 20 packets.
b Find the variance of black jelly beans selected from 50 packets.

$E(X) = 20(0.16) = 3.2$

$\text{Var}(X) = 50(0.16)(0.84) = 6.72$

7 Silvana rolls a die 50 times.

a What is the expected number of sixes that she rolls?

$E(X) = 50 \times \frac{1}{6} = 8.33\ldots$

8 sixes

b Find the probability of Silvana rolling this number of sixes.
Leave your answer in index form.

$P(X = 8) = \binom{50}{8}\left(\frac{1}{6}\right)^{8}\left(\frac{5}{6}\right)^{42}$

8 Faisal takes the 4 aces from a deck of cards and randomly selects an ace out of these 4 cards over 100 experiments.

a What is the most likely number of times that he selects the ace of hearts?

$E(X) = 100 \times \frac{1}{4}$

25 hearts

b Find the probability that he selects the ace of hearts this number of times, to 3 s.f.

$P(X = 25) = \binom{100}{25}\left(\frac{1}{4}\right)^{25}\left(\frac{3}{4}\right)^{75} = 0.0918$

9

a When 2 dice are rolled, find the probability that the difference of the numbers on the dice is 2.

The favourable ordered pairs are $(1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (6,4)$.

$P(\text{difference } 2) = \frac{8}{36} = \frac{2}{9}$

b Two dice are rolled 50 times. For the binomial distribution of rolling a difference of 2, find:

i the mean

ii the variance

$X \sim \text{Bin}\left(50, \frac{2}{9}\right)$

$E(X) = 50\left(\frac{2}{9}\right) = 11\frac{1}{9}$

$\text{Var}(X) = 50\left(\frac{2}{9}\right)\left(\frac{7}{9}\right) = 8\frac{52}{81}$

10 A binomial distribution has mean 68 and variance 13.6. Write the distribution as a table.

Using $np = 68$ and $np(1-p) = 13.6$:

$1 - p = \frac{13.6}{68} = \frac{1}{5}$

$p = \frac{4}{5}$

$n = \frac{68}{4/5} = 85$

Therefore $X \sim \text{Bin}\left(85, \frac{4}{5}\right)$

Underlying trial distribution:

X	0	1
$P(X = x)$	$\frac{1}{5}$	$\frac{4}{5}$

11 At the school fete, few people have entered any of the 24 meat raffles, so Larry buys 8 of the 20 tickets in each raffle.

a Calculate the expected value and standard deviation for the random variable of the number of Larry's wins.

b If Larry wins 6 raffles, calculate how many standard deviations Larry's result is below the mean, as a measure of his poor luck.

Let $X \sim B(24, \frac{8}{20}) = B(24, 0.4)$.

$E(X) = np = 24(0.4) = 9.6$

$\sigma = \sqrt{npq} = \sqrt{24(0.4)(0.6)} = 2.4$

$z = \frac{6 - 9.6}{2.4} = -1.5$

Larry's result is 1.5 standard deviations below the mean.

12

a If six dice are rolled one hundred times, how many times would you expect the number of even numbers showing to exceed the number of odd numbers showing?

b If eight coins are tossed sixty times, how many times would you expect the number of heads to exceed the number of tails?

For one set of six dice, let $X \sim B(6, \frac{1}{2})$ record even results.

$P(X > 3) = \frac{\binom{6}{4} + \binom{6}{5} + \binom{6}{6}}{2^6} = \frac{22}{64}$

Expected number $= 100\left(\frac{22}{64}\right) = 34.375 \approx 34$ times.

For one set of eight coins, let $X \sim B(8, \frac{1}{2})$ record heads.

$P(X > 4) = \frac{\binom{8}{5} + \binom{8}{6} + \binom{8}{7} + \binom{8}{8}}{2^8} = \frac{93}{256}$

Expected number $= 60\left(\frac{93}{256}\right) = 21.80 \approx 22$ times.

13

a If a player throws 6 dice, what is the probability of getting at least two sixes? Let the random variable X record the number of sixes. What is the expected value and standard deviation for X?

Let $X \sim B(6, \frac{1}{6})$.

$P(X \geq 2) = 1 - \left(\frac{5}{6}\right)^6 - 6\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^5 = 0.263 \approx 26\%$

$E(X) = np = 1$

$\sigma = \sqrt{npq} = \sqrt{\frac{5}{6}} \approx 0.91$

b Repeat the question for 24 dice.

For $n = 24$:

$P(X \geq 2) = 1 - \left(\frac{5}{6}\right)^{24} - 24\left(\frac{1}{6}\right)\left(\frac{5}{6}\right)^{23} = 0.927 \approx 93\%$

$E(X) = 4, \sigma = \sqrt{\frac{10}{3}} \approx 1.83.$

14 In a simple experiment, 6 fair coins are tossed and the number of heads is recorded. A student wishes to compare the results with theoretical predictions.

a Complete the table below using the formula $P(X = x) = \binom{n}{x}p^x q^{n-x}$ for binomial probability.

Number of heads	0	1	2	3	4	5	6	Total
Number of ways								
Probability								

Heads x	0	1	2	3	4	5	6	Total
Ways	1	6	15	20	15	6	1	64
$P(X = x)$	$\frac{1}{64}$	$\frac{6}{64}$	$\frac{15}{64}$	$\frac{20}{64}$	$\frac{15}{64}$	$\frac{6}{64}$	$\frac{1}{64}$	1

b Read off the mode, the most common result.

The mode is 3 heads.

c Use this table to determine the expected value and variance of this discrete probability distribution.

$$E(X) = \sum xP(X = x) = 3$$

$$E(X^2) = \sum x^2P(X = x) = 10.5$$

$$\text{Var}(X) = E(X^2) - [E(X)]^2 = 10.5 - 9 = 1.5$$

d Compare your results to those obtained using $E(X) = np$ and $\text{Var}(X) = npq$, where $q = 1 - p$.

$$E(X) = np = 6\left(\frac{1}{2}\right) = 3$$

$$\text{Var}(X) = npq = 6\left(\frac{1}{2}\right)\left(\frac{1}{2}\right) = 1.5$$

The results agree.

e Explain, in your own words, why the result for the expected value is not a surprise.

The distribution is symmetric, so its mean is at the midpoint, $x = 3$.

15 For each situation, construct a table for the theoretical distribution associated with the given random variable and calculate the mode, mean and standard deviation.

a A coin is tossed five times and X is the number of heads that are face up.

$X \sim B(5, \frac{1}{2})$.

x	0	1	2	3	4	5
$P(X = x)$	$\frac{1}{32}$	$\frac{5}{32}$	$\frac{10}{32}$	$\frac{10}{32}$	$\frac{5}{32}$	$\frac{1}{32}$

Mode $= 2$ or 3.

$\mu = np = \frac{5}{2}$

$\sigma = \sqrt{npq} = \frac{\sqrt{5}}{2}$

b A die is thrown five times and X is the number of times 5 or 6 occurs.

$X \sim B(5, \frac{1}{3})$.

x	0	1	2	3	4	5
$P(X = x)$	$\frac{32}{243}$	$\frac{80}{243}$	$\frac{80}{243}$	$\frac{40}{243}$	$\frac{10}{243}$	$\frac{1}{243}$

Mode $= 1$ or 2.

$\mu = np = \frac{5}{3}$

$\sigma = \sqrt{npq} = \frac{\sqrt{10}}{3}$

c Five cards are drawn in turn with replacement from a standard pack of 52 cards and X is the number of court cards, jack, queen or king, that are drawn. Give your result correct to 3 decimal places.

A court card has probability $p = \frac{12}{52} = \frac{3}{13}$.

x	0	1	2	3	4	5
$P(X = x)$	0.269	0.404	0.242	0.073	0.011	0.001

Mode $= 1$.

$\mu = np = \frac{15}{13} \approx 1.15$

$\sigma = \sqrt{npq} = \frac{\sqrt{150}}{13} \approx 0.94$

16 It is known that 15% of a large constituency voted for the Working Together Party at the last election. A poll of 100 people is taken and they are asked whether they voted WTP.

a What is the mean and standard deviation for this poll?

b What is the probability that in the sample, the number of WTP voters lies within half a standard deviation of the mean?

Let $X \sim B(100, 0.15)$.

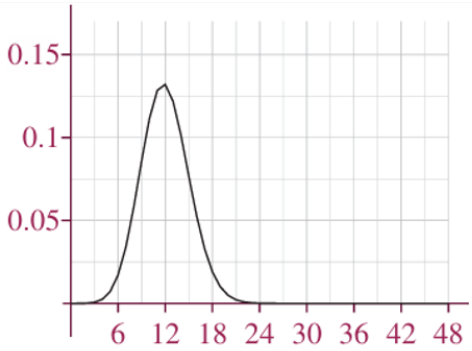
$\mu = np = 15$

$\sigma = \sqrt{npq} = \sqrt{100(0.15)(0.85)} \approx 3.57$

Half a standard deviation gives $15 \pm \frac{3.57}{2}$, so the integer values are 14, 15 and 16.

$$P(14 \leq X \leq 16) = \sum_{x=14}^{16} \binom{100}{x} (0.15)^x (0.85)^{100-x} \approx 0.325$$

17 A relative-frequency polygon is shown for the binomial distribution with $n = 48$ and $p = 0.25$.



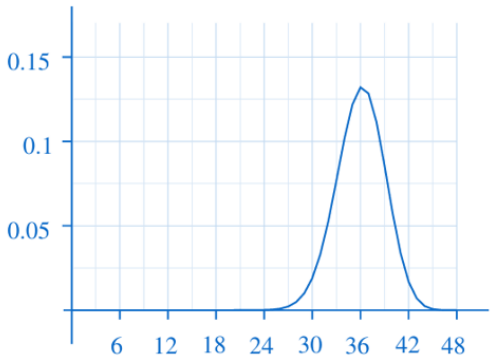
a Is the distribution skewed to the left, skewed to the right, or not at all?

b Find the mean and standard deviation for the distribution.

c What is the mode of the distribution? Estimate the probability of this most likely outcome.

d Shade the region no more than two standard deviations from the mean.

e Use the given sketch for $p = 0.25$ to assist in sketching the graph of the distribution for $p = 0.75$.



Since $p < 0.5$, the distribution is skewed to the right.

$\mu = np = 48(0.25) = 12$

$\sigma = \sqrt{npq} = \sqrt{48(0.25)(0.75)} = 3$

The mode is 12.

$P(X = 12) = \binom{48}{12}(0.25)^{12}(0.75)^{36} \approx 0.13$

Two standard deviations from the mean gives:

$12 - 2(3) \leq x \leq 12 + 2(3)$

Shade $6 \leq x \leq 18$.

The $p = 0.75$ graph is the $p = 0.25$ graph reflected horizontally in $x = 24$.

This is due to the identity $\binom{n}{k} = \binom{n}{n-k}$

The distribution for $p = 0.75$ is the number of failures for the same experiment, so it has the exact same shape.

$k \mapsto 48 - k$, this is a horizontal shift left by 48, then a reflection across y-axis, which is equivalent to reflecting across $x = 24$
(note: reflection across $x = a$ is not in syllabus)

18 A certain drug is found to be helpful for 70% of patients who take it. Two research teams are attempting to improve its effectiveness by adjusting the delivery system for the drug.
Both conduct random trials to test the effectiveness of their changes.
Team A runs a trial with 50 patients, of whom 45 show improvement using the drug.
Team B runs a trial with 90 patients, of whom 74 show improvement.
Use the mean and standard deviation of the two trials to decide which adjustment shows stronger evidence of an improvement.

Team A: $X_A \sim B(50,0.7)$.

$\mu_A = 35$, $\sigma_A = \sqrt{50(0.7)(0.3)} \approx 3.2$

$$z_A = \frac{45 - 35}{3.2} \approx 3.1$$

Team B: $X_B \sim B(90,0.7)$.

$\mu_B = 63$, $\sigma_B = \sqrt{90(0.7)(0.3)} \approx 4.3$

$$z_B = \frac{74 - 63}{4.3} \approx 2.6$$

Team A's result is further above its mean, so Team A shows stronger evidence of improvement. Further trials are needed.

19 Ms Taylor sets her class a test of 48 multiple-choice questions, each with four options A-D. Let the random variable X be the number of questions a person gets correct.

a Calculate the expected value $E(X)$. How can this value be understood in this context?

b What is the standard deviation σ of the random variable?

c Ms Taylor is annoyed to discover that Fayola, one of her students, claims to have got 24 just by guessing. How many standard deviations is Fayola's score above the mean?

For random guessing on the first test, $X \sim B(48,0.25)$.

$E(X) = np = 48(0.25) = 12$

A student guessing randomly would average 12 correct answers.

$\sigma = \sqrt{48(0.25)(0.75)} = 3$

$z = \frac{24 - 12}{3} = 4$

Fayola's score is 4 standard deviations above the mean.

For the second test, $Y \sim B(100,0.2)$.

Ms Taylor writes a new test, with 100 questions, each with five options A-E.

d What is the expected value and standard deviation of this new distribution with random variable Y?

e Fayola gets 40 questions right this time, and again she claims to have achieved this result just by guessing. If true, would this be more or less unusual than her previous result?

f Idette gets 75% in the first test and 60% in the second. Which is the more unusual of her results, if she is guessing?

$E(Y) = 100(0.2) = 20$

$\sigma = \sqrt{100(0.2)(0.8)} = 4$

$z = \frac{40 - 20}{4} = 5$

This result is more unusual than her first result.

First test: 75% of 48 is 36.

$z = \frac{36 - 12}{3} = 8$

Second test: 60% of 100 is 60.

$z = \frac{60 - 20}{4} = 10$

The second result is more unusual.